

Finlay Cooper

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Education

Western University, London ON

June 2026

Bachelor of Engineering, Software Engineering | **Dean's Honour List**

- **Core Coursework:** Algorithms & Data Structures; Operating Systems; DBMS; Networking; Web Technologies; Information Security; Machine Learning; Cloud Computing; Software Testing & Maintenance.

Experience

Software Developer Intern, BroadGrain Commodities — Toronto, ON

Summer 2025

- **Owned full product lifecycle** as sole developer, defining requirements with plant managers, managing the backlog, and shipping iterative releases under real operational deadlines.
- Engineered a **schema-driven backend factory** to auto-generate CRUD controllers, routes, and validation from a single schema, reducing 100+ files to 3 and enabling zero-code endpoint creation.
- Integrated **Microsoft Graph API + Azure AD** to automate SharePoint document delivery with token caching, retries, and exponential backoff, eliminating manual report distribution entirely.
- Built **10 PDF reporting services** (JSReport, Handlebars, digital signatures, multi-photo support), replacing paper-based compliance workflows across 8 categories.
- Integrated **LLM tooling (MCP)** for automated legacy data migration and schema derivation; deployed to on-prem infrastructure (**pm2/IIS**) and authored runbooks for successor handoff.
- **Reduced processing time by 15% (refactoring) and 75% (digitization)** across reporting workflows.

Projects

Fracture Protocol — 2-Person Unity Roguelike

2025–2026

- Engineered a **top-down roguelike arena shooter** in Unity/C# with 77 scripts, 3 boss encounters, 12 elite enemy archetypes, and a **dual-layer progression system** (in-run modifiers + permanent meta upgrades).
- Designed and implemented **coroutine-driven boss AI** for 3 distinct encounters, running concurrent behaviors (fire patterns, drone spawning phases, respawn timers) simultaneously with phase transitions at 66%/33% HP.
- Built a **32-modifier system** with 5 rarity tiers, stack-capped diminishing returns, and a central string-ID dispatch queried at runtime—enabling dynamic effect composition without hardcoded per-modifier branches.
- Engineered **wave-scaled enemy HP** ($100 + \min(50, (\text{wave}-1)*0.85)$) and a curated elite schedule across 55+ waves, balancing early-game clarity with late-game challenge without creating bullet sponges.
- Coordinated **Git version control** across concurrent Unity development—managed branching, resolved prefab and scene merge conflicts, and maintained stable builds across 2 active contributors.

Quizzly — Real-Time Multiplayer Trivia Game | github.com/FinlayCooper/Quizzly

Jan–Apr 2025

- Built a **real-time multiplayer trivia game** with a C++ backend managing concurrent game sessions and player connections via **WebSockets** for low-latency synchronous gameplay.
- Implemented **multithreaded session management** to handle simultaneous game rooms, with a custom networking protocol for reliable message passing and game state synchronization across clients.
- Collaborated across **5 contributors** using Git, managing concurrent development workflows and version control on a shared codebase.

Against the Machines — 3D Unity Action Game | Against-the-Machines

Fall 2023

- Built a **three-level Unity/C# action game**; implemented player controls, enemy AI, and physics-driven interactions.
- Delivered across **three Agile sprints** with weekly stand-ups, demos, and retrospectives.

Skills

Languages & Frameworks: C#, C++, Python, TypeScript/JavaScript, Java, SQL; Unity, Swift/SwiftUI, React/React Native, Next.js, Node.js, Express.js

Tools & Platforms: Git, Docker, CI/CD; MySQL, PostgreSQL, Firestore, MongoDB; Prisma ORM; Azure/SharePoint, IIS, Nginx; Networking (IP, Routing & Switching)

Spoken Languages: English, French